

ASVAB AFQT Math Cheat-Sheet

Every formula you need for Arithmetic Reasoning (AR) & Mathematics Knowledge (MK) — the math half of your AFQT score.

The AFQT (your enlistment-qualifying score) is half math. Most of it isn't hard computation — it's **knowing the right setup**. Keep this sheet next to you while you practice. ⚠ = the mistake that trips most test-takers.

PART 1 • Arithmetic Reasoning (word problems)

Arithmetic Fundamentals

- ▶ Order of operations (PEMDAS): Parentheses → Exponents → Multiply/Divide (L→R) → Add/Subtract (L→R)
- ▶ Even $\times$ anything = even; odd $\times$ odd = odd
- ▶ Negative $\times$ negative = positive; negative $\times$ positive = negative
- ▶ Divisible by 3 if the digit sum is divisible by 3 (123 $\rightarrow$ 1+2+3 = 6 ✓)
- ▶ Remainder: $a \div b$ leaves r means $a = b \times \text{quotient} + r$

⚠ **Trap:** Doing multiplication before resolving parentheses.

Averages & Mean

- ▶ Mean = sum of values $\div$ number of values
- ▶ Sum = average $\times$ number of values
- ▶ Missing value for a target: needed = (target avg $\times$ count) – (sum so far)
- ▶ Weighted average = total of (value $\times$ weight) $\div$ total weight
- ▶ Average speed (equal distances) = total distance $\div$ total time

⚠ **Trap:** Averaging two speeds directly instead of total distance $\div$ total time.

Percent & Markup

- ▶ Part = (percent $\div$ 100) $\times$ whole
- ▶ What percent: percent = (part $\div$ whole) $\times$ 100
- ▶ Find the original: whole = part $\div$ (percent $\div$ 100)
- ▶ % change = (new – old) $\div$ old $\times$ 100
- ▶ Markup: new = original $\times$ (1 + rate). Discount: sale = original $\times$ (1 – rate)

⚠ **Trap:** Percent change is measured off the ORIGINAL, not the new value.

Rate, Distance & Time

- ▶ Distance = Rate $\times$ Time ($D = R \times T$)
- ▶ $R = D \div T$ • $T = D \div R$
- ▶ Round-trip average speed = total distance $\div$ total time
- ▶ Combined rate (workers/pipes) = rate₁ + rate₂

⚠ **Trap:** Mixing units — convert minutes to hours before using mph.

Ratios & Proportions

- ▶ Proportion $a/b = c/d \rightarrow$ cross-multiply: $a \times d = b \times c$
- ▶ Unit rate = quantity $\div$ count (\$4.50 $\div$ 3 = \$1.50 each)
- ▶ Scale up: unit rate $\times$ new count
- ▶ Part-to-whole: ratio A:B means A out of every (A+B)

⚠ **Trap:** Confusing 'ratio of A to B' (A/B) with 'A of the total' (A/(A+B)).

Work & Rate

- ▶ One worker's rate = 1 job $\div$ solo time (5 hrs $\rightarrow$ rate 1/5 per hr)
- ▶ Combined: $1/a + 1/b = 1/t$
- ▶ Time together (two workers) = $(a \times b) \div (a + b)$
- ▶ Fill & drain: net rate = fill – drain, then time = $1 \div$ net rate

⚠ **Trap:** Adding the times instead of the rates (two 6-hr workers finish in 3 hrs, not 12).

Multi-Step Word Problems

- ▶ Read the LAST sentence first — solve for what's actually asked
- ▶ Label every number with its unit to catch setup errors
- ▶ Estimate with round numbers to sanity-check the magnitude
- ▶ Not every number is needed — some are distractors

⚠ **Trap:** Answering the wrong question. Re-read what it asks before you pick.

PART 2 • Mathematics Knowledge (concepts)

Linear Equations & Algebra

- ▶ Solve for x: do the same operation to BOTH sides
- ▶ Distribute: $a(b + c) = ab + ac$
- ▶ Combine like terms: $3x + 5x = 8x$ (same variable only)
- ▶ Two-step: undo $+/-$ first, then $\times/\div$
- ▶ Slope-intercept: $y = mx + b$ (m = slope, b = y-intercept)

⚠ **Trap:** Sign error distributing a negative: $-2(x - 3) = -2x + 6$, not $-2x - 6$.

Exponents & Polynomials

- ▶ Product: $x^a \times x^b = x^{a+b}$ (add exponents)
- ▶ Quotient: $x^a \div x^b = x^{a-b}$ (subtract)
- ▶ Power: $(x^a)^b = x^{ab}$ (multiply)
- ▶ $x^0 = 1$ (any nonzero x) • $x^{-a} = 1/x^a$
- ▶ $(xy)^a = x^a \times y^a$

⚠ **Trap:** $x^2 \times x^3 = x^5$, NOT x^6 — you add the exponents, not the bases.

Fractions, Decimals & Percent

- ▶ Fraction $\rightarrow$ decimal: divide top by bottom ($3/4 = 0.75$)
- ▶ Decimal $\rightarrow$ percent: $\times 100$ • percent $\rightarrow$ decimal: $\div 100$
- ▶ Add/subtract: need a common denominator first
- ▶ Multiply: straight across, then simplify
- ▶ Divide: flip the second fraction and multiply

⚠ **Trap:** Adding denominators ($1/3 + 1/4 \neq 2/7$) — find a common denominator.

Geometry: Area, Perimeter & Volume

- ▶ Rectangle: $A = l \times w$; $P = 2(l + w)$
- ▶ Triangle: $A = \frac{1}{2} \times \text{base} \times \text{height}$
- ▶ Circle: $A = \pi r^2$; $C = 2\pi r$ ($\pi \approx 3.14$)
- ▶ Trapezoid: $A = \frac{1}{2} \times (b_1 + b_2) \times \text{height}$
- ▶ Box: $V = l \times w \times h$ • Cylinder: $V = \pi r^2 \times h$

⚠ **Trap:** Using diameter where the formula needs radius ($r = \text{diameter} \div 2$).

Inequalities

- ▶ $<$ less than, $>$ greater, $\leq/\geq$ or-equal-to
- ▶ Add/subtract the same amount: symbol stays the same
- ▶ $\times$ or $\div$ by a POSITIVE: symbol stays
- ▶ $\times$ or $\div$ by a NEGATIVE: FLIP the symbol
- ▶ Compound: $a < x < b$ means x is between a and b

⚠ **Trap:** Forgetting to flip the symbol after dividing by a negative.

Number Properties & Absolute Value

- ▶ $|x|$ is always non-negative ($|-7| = 7$)
- ▶ even+even=even; odd+odd=even; even+odd=odd
- ▶ neg $\times$ neg = pos; opposite signs = neg
- ▶ Primes: 2, 3, 5, 7, 11, 13, 17, 19 (1 is NOT prime)
- ▶ GCF = largest shared factor; LCM = smallest shared multiple

⚠ **Trap:** Thinking 1 is prime — it isn't (it has only one factor).

Order of Operations (PEMDAS)

- ▶ Parentheses $\rightarrow$ Exponents $\rightarrow$ Mult/Div $\rightarrow$ Add/Sub
- ▶ Multiply & divide share one rank: go left to right
- ▶ Add & subtract share one rank: go left to right
- ▶ Exponents before multiplication: $2 \times 3^2 = 2 \times 9 = 18$
- ▶ Distribute a negative: $-(a - b) = -a + b$

⚠ **Trap:** $2 + 3 \times 4 = 14$ (multiply first), not 20.